University of Information Technology & Sciences (UITS)

Faculty of Science and Engineering

Department of Computer Science and Engineering

Program: B.Sc. in CSE

Lab Test 01, Spring 2026

Course Title: Microprocessors & Microcontrollers Lab

Course Code: CSE0611323

Set A

Marks: 10 **Section: 6B** **Time: 1hr**

(Answer all questions)

Instructions:

1. Write your codes using standard 8086 Assembly Language (compatible with EMU8086).
2. Add appropriate comments to explain your logic. Marks will be deducted for uncommented code.
3. The output should be according to the sample output. If sample output is not followed then marks will be deducted.

Questions

1. Write assembly code that reads alphabetic characters from the keyboard.
2. Convert the character to its opposite case (if it was uppercase, convert it to lowercase; if it was lowercase, convert it to uppercase) and display the converted character on the next line.
3. Finally, perform a rotate left operation on the converted character and print the Hex code.
4. Print as per the sample output

Sample **Output:**

Input = A
Converted Character = a
Hex code after rotate left = C2H

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Lab Test 01, Autumn 2026

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Set B

Marks: 10 | **Section: 6B** | **Time: 1hr**

(Answer all questions)

Instructions:

1. Write your codes using standard 8086 Assembly Language (compatible with EMU8086).
2. Add appropriate comments to explain your logic. Marks will be deducted for uncommented code.
3. The output should be according to the sample output. If sample output is not followed then marks will be deducted.

Questions

1. Write assembly code that reads alphabetic characters from the keyboard.
2. Convert the character to its opposite case (if it was uppercase, convert it to lowercase; if it was lowercase, convert it to uppercase) and display the converted character on the next line.
3. Finally, perform a shift right operation on the converted character and print the Hex code.
4. Print as per the sample output.

Sample Output:

Input = A
Converted Character = a
Hex code after shift right = 30H